5.7 Industry Core Setup - Example

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Authors:	Markus Keidl (FG-233) Christopher Winter (FG-233)

Change History

Version	Date	Authors	Changes
1.0.0	26.11.2024	Markus Keidl (FG-233) Christopher Winter (FG-233)	Release of initial version of the Digital Twin Example.

The complete example is available for testing in the following BMW EDC:

- EDC URL: https://connector-partner.edc.aws.bmw.cloud/api/v1/dsp
- BPN: BPNL0000000J8CWU

Please be aware that every partner that wants to test needs to be added to the EDC access policy BPN group first. For that, write an email with the partner's BPN to Markus Keidl (FG-233) and Christopher Winter (FG-233) .

Additional documentation is available here:

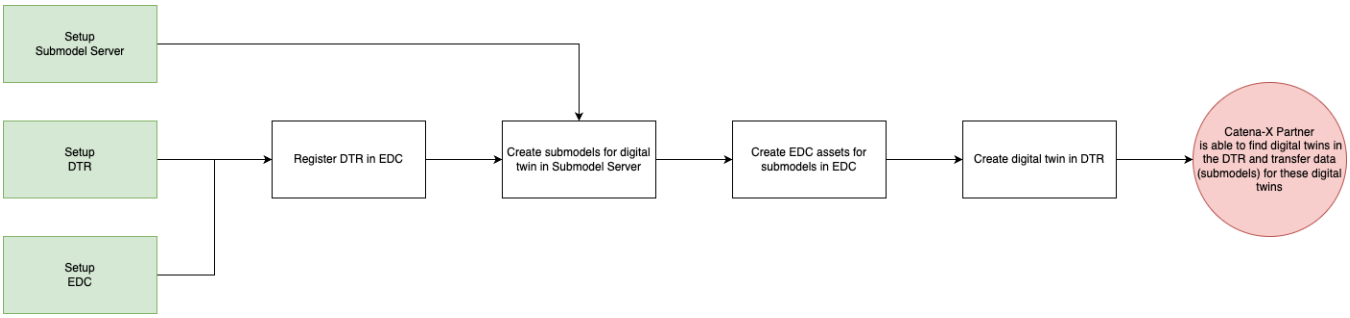
- BMW-internal GitHub repository with all files used in the example: https://atc-github.azure.cloud.bmw/Extended-Enterprise-Catena-X/catena-x-dt-example
- Insomnia collection with all request to setup the example or query DTR and submodel data: Digital-Twin-Example-v1-0.0.json

Introduction to the Example

This example shows you how to create digital twins and provide these in Catana-X in a Industry-Core-compliant way. You need to setup the following components to implement this example:

- Digital Twin Registry (DTR) - to create and publish digital twins
- Submodel Server - to create and provide data (so-called submodels) for digital twins based on aspect models
- EDC - to connect DTR and Submodel Server to Catena-X. Both components, DTR and Submodel Server are only accessible via the EDC.

You need all of these three components and you need to connect them properly to that data consumers can find your digital twins and transfer data (so-called submodels) for these digital twins. The following steps are necessary to successfully setup this example:

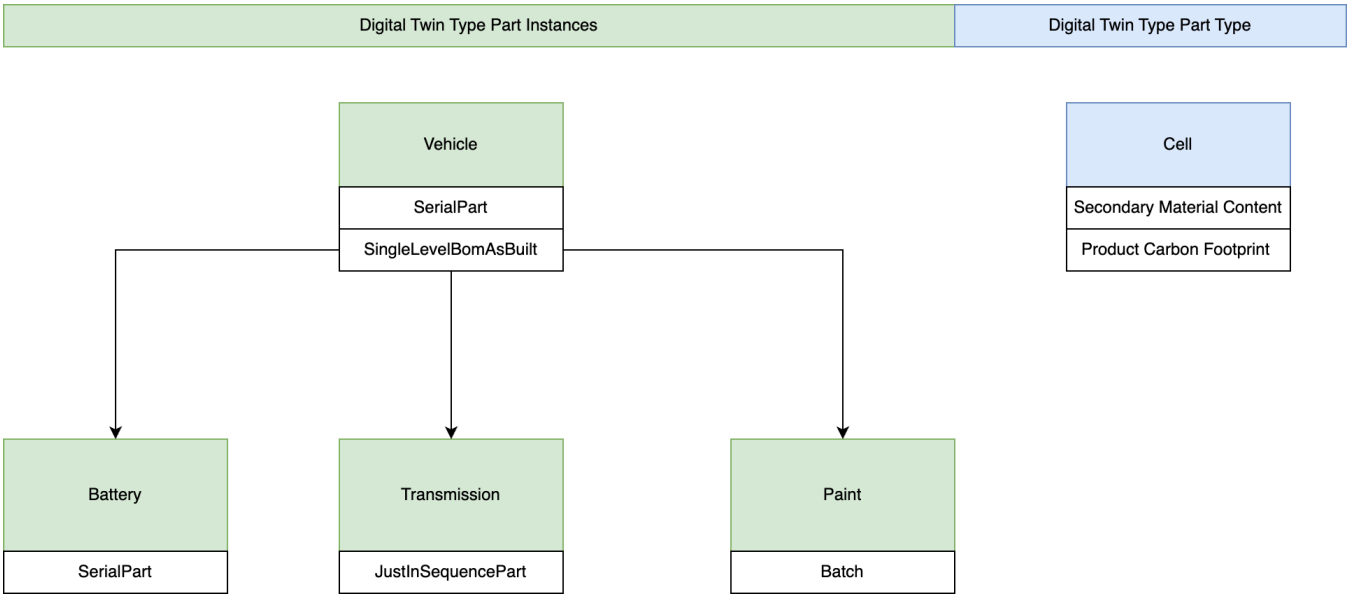


The example consists of

- PartInstance: Vehicle with three components Battery (Serial Part), Transmission (JISPart), Paint (Batch)
  - For the Vehicle digital twin, we create two submodels: SerialPart and SingleLevelBomAsBuilt
  - The BoM of the Vehicle, contains three child parts: a Battery, a Transmission, and Paint. The SingleLevelBomAsBuilt submodel contains this information.
  - For the Battery, Transmission, and Paint digital twins, we only create one submodel: SerialPart, JustInSequencePart, or Batch, depending on the type of part.
  - To publish the data in the EDC we use two approaches which are also described in the Industry Core KIT. More details can be found in the following sections.
    - No asset bundling is used for Vehicle and Transmission digital twins
    - Asset bundling is used for Battery digital twins.
- PartType:
  - 1 Twin for CellType with submodels SRQ, PCF

We also provide an Insomnia collection with all steps necessary to create DTR and EDC artifacts for this example. The Submodel Server uses a simple S3 bucket for data for non-asset-bundling submodels and a simple HTTP server for asset-bundling submodels.

The following figure illustrates the example:



Digital Twin	ID in DTR (Property "id")	Submodels	Asset ID in EDC	Asset Bundling	Comment
BMW i4 2024-01-01-0001 (Vehicle)	urn:uuid:da071e28-8cf2-46a1-b5af-65231887c7c5				You can use the following query to fetch the digital twin from the DTR (by ID): https://registry-partner.edc.aws.bmw.cloud/api/v3/shell-descriptors /dXJuOnV1aWQ6ZGEwNzFlMjgtOGNmMi00NmExLWI1YWYtNjUyMzE4ODdjN2M1 dXJuOnV1aWQ6ZGEwNzFlMjgtOGNmMi00NmExLWI1YWYtNjUyMzE4ODdjN2M1 is the base64-encoded digital twin ID urn:uuid:da071e28-8cf2-46a1-b5af-65231887c7c5

		SerialPart	Vehicle_2024-01-01-0001_Submodel_SerialPart	No	
		SingleLevelBomAsBuilt	Vehicle_2024-01-01-0001_Submodel_SingleLevelBomAsBuilt	No	
Battery 2024-01-01-0001	urn:uuid:7cf7edda-1efa-459b-b9d7-7e9f29d8d199				
		SerialPart	Battery_Submodel_SerialPart	Yes	
Transmission 2024-01-01-0001	urn:uuid:57d081ea-9d25-4ac5-ada0-7deb0c6b1f07				
		JustInSquence Part	Transmission_2024-01-01-0001_Submodel_JustInSequencePart	No	
Paint					Not yet implemented
LI-ION ZELLE PHEV1 26AH MK02 C2	`urn:uuid:5d63f272-8029-4ee6-8e3d-3b1831a30b36`				
		PCF	Submodel_Cell_ProductCarbonFootprint		
		SMC	Submodel_Cell_SecondaryMaterialContent		

Setup DTR in EDC

Create DTR Asset

In a first step a **EDC asset** for the **decentral Digital Twin Registry** needs to be created. Therefore the following json can be used to create your dDTR asset:

dDTR Asset

```
{
  "@context": {
    "edc": "https://w3id.org/edc/v0.0.1/ns/",
    "cx-common": "https://w3id.org/catenax/ontology/common#",
    "cx-taxo": "https://w3id.org/catenax/taxonomy#",
    "dct": "http://purl.org/dc/terms/"
  },
  "@id": "{{ _.edcAssetId }}",
  "properties": {
    "dct:type": {
      "@id": "cx-taxo:DigitalTwinRegistry"
    },
    "cx-common:version": "3.0"
  },
  "privateProperties": {
  },
  "dataAddress": {
    "@type": "DataAddress",
    "type": "HttpData",
    "baseUrl": "{{DTR_URL}}",
    "proxyQueryParams": "true",
    "proxyPath": "true",
    "proxyMethod": "false",
    "oauth2:tokenUrl": "{{ _.url_keycl_backend }}",
    "oauth2:clientId": "{{ _.client_id_backend }}",
    "oauth2:clientSecretKey": "{{ _.sec_name_vault }}"
  }
}
```

- **Asset ID** can me free chosen. We recommend to use a **UUID v4**
- The **dct:type** MUST be set to cx-taxo:DgitalTwinRegistry"
- The **baseUrl** end before the /shell-descriptor and /lookup segments
- The **proxyPath** can me set to true or false. If true the data plan of the edc will forward the request to the DTR.
- We recommend to ser the **proxyMehtod** to **false** as CX currently not forsee to allow 3th parties to change dDTR entries.

As BMW we search for the DTR in the edc catalogue using the **following filter**:

dDTR Asset

```
{
  "@context": {
    "dct": "http://purl.org/dc/terms/"
  },
  "protocol": "dataspace-protocol-http",
  "counterPartyAddress": "DATAPROVIDERURL/api/v1/dsp",
  "counterPartyId": "PARTNERURL",
  "querySpec": {
    "@type": "QuerySpecDto",
    "https://w3id.org/edc/v0.0.1/ns/offset": 0,
    "https://w3id.org/edc/v0.0.1/ns/limit": 10,
    "https://w3id.org/edc/v0.0.1/ns/filterExpression": {
      "@type": "CriterionDto",
      "operandLeft": "http://purl.org/dc/terms/type",
      "operator": "like",
      "operandRight": "%https://w3id.org/catenax/taxonomy#DigitalTwinRegistry%"
    }
  }
}
```

Creating a dDTR Contract Definition

Info

In common a contract definition always consists out of 1 to n assets an access policy and 1 to n usage policies. So you should first create a access and usage policy for your registry. in common policies can be reused, but please **do not reuse the usage policy** of the registry for other contract definitions as the purpose of the usage is focused on the dDTR use!

Access Policy for dDTR

dDTR Asset

```
{
        "@id": "ACCESSPOLICYIDfordDTR",
        "@type": "PolicyDefinition",
        "policy": {
                "@type": "odrl:Set",
                "odrl:permission": {
                        "odrl:action": {
                                "@id": "odrl:use"
                        },
                        "odrl:constraint": {
                                "odrl:and": [
                                        {
                                                "odrl:leftOperand": {
                                                        "@id": "cx-policy:Membership"
                                                },
                                                "odrl:operator": {
                                                        "@id": "odrl:eq"
                                                },
                                                "odrl:rightOperand": "active"
                                        },
                                        {
                                                "odrl:leftOperand": {
                                                        "@id": "tx:BusinessPartnerGroup"
                                                },
                                                "odrl:operator": {
                                                        "@id": "odrl:isAnyOf"
                                                },
                                                "odrl:rightOperand": "digital-twin-exchange-example"
                                        }
                                ]
                        }
                },
                "odrl:prohibition": [],
                "odrl:obligation": []
        },
        "@context": {
                "@vocab": "https://w3id.org/edc/v0.0.1/ns/",
                "edc": "https://w3id.org/edc/v0.0.1/ns/",
                "tx": "https://w3id.org/tractusx/v0.0.1/ns/",
                "tx-auth": "https://w3id.org/tractusx/auth/",
                "cx-policy": "https://w3id.org/catenax/policy/",
                "odrl": "http://www.w3.org/ns/odrl/2/"
        }
}
```

Usage Policy for dDTR

In teh next step please create a usage policy specific for your Digital Twin

dDTR Asset

```
{
  "@context": {
    "@vocab": "https://w3id.org/edc/v0.0.1/ns/",
    "edc": "https://w3id.org/edc/v0.0.1/ns/",
    "cx-policy": "https://w3id.org/catenax/policy/",
    "odrl": "http://www.w3.org/ns/odrl/2/"
  },
  "@id": "USEAGEPOLICYIDfordDTR",
  "policy": {
    "@type": "odrl:Set",
    "odrl:permission": {
      "odrl:action": {
        "@id": "odrl:use"
      },
      "odrl:constraint": {
        "odrl:and": [
          {
            "odrl:leftOperand": {
              "@id": "cx-policy:FrameworkAgreement"
            },
            "odrl:operator": {
              "@id": "odrl:eq"
            },
            "odrl:rightOperand": "DataExchangeGovernance:1.0"
          },
          {
            "odrl:leftOperand": {
              "@id": "cx-policy:UsagePurpose"
            },
            "odrl:operator": {
              "@id": "odrl:eq"
            },
            "odrl:rightOperand": "cx.core.digitalTwinRegistry:1"
          }
        ]
      }
    }
  }
}
```

Register Digital Twins in DTR

Registering digital twins in the DTR is straight forward. For convenience for this example, we register every digital twin in one step, i.e., including all submodel descriptors. The DTR API also allows you to create the digital twin in multiple steps. Some additional explanatory notes:

- The property `idShort` of a digital twin entry must be unique. So we added a date counter (e.g., "Battery 2024-01-01-0001" instead of just "Battery").
- The visibility of certain attributes must be restricted. For example, not all specific asset IDs of a digital twin are allowed to be public readable. For this example, we still made all IDs public. Otherwise, we would have needed to add visibility permissions to the digital twin whenever a new partner wants to access the example. To limit visibility, you would use the property `externalSubjectId`. Here's an example that grants the Catena-X partner with BPN <PARTNER_BPN> access to the specific asset ID partInstanceId:

```
{
        "name": "partInstanceId",
        "value": "T588540140123B1011B018362515271",
        "externalSubjectId": {
        "type": "ExternalReference",
        "keys": [
                {
                        "type": "GlobalReference",
                        "value": "<PARTNER_BPN>"
                }
        ]
}
```

More details about digital twin visibility can be found in the CX-0002 Digital Twins in Catena-X standard and the Digital Twin KIT.

Registering the Vehicle Digital Twin (Part Instance)

The following example shows the full digital for a Vehicle including submodel descriptors for SerialPart and SingelLevelBomAsBuilt submodels. This digital twin is registered using the method POST /shell-descriptors of the DTR API.

- Mandatory and optional specific asset IDs depend on the actual type of the part for which the digital twin is created: serial part, batch, or Just-In-Sequence part. More details can be found here. Vehicles are serial parts in this context as they are identified using a serial number (VIN or - in pseudonymized form - VAN).
- For every submodel descriptor, the aspect model which defines the schema of the submodel must be specified using the `semanticId` property.
- Property `subprotocolBody` where the submodel data can be found: `dspEndpoint` points to the EDC (more accurately, to the catalog service of the EDC), `id` points to the EDC asset from which the submodel can be transferred.
- Once the data consumer, successfully negotiated a contract for this EDC asset, the URL in the `href` property must be used to actually transfer the data.
- Depending on the approach how submodels are published in the EDC (with or without asset bundling), these properties will look different. The sections below give more details about this.

AAS descirptor + Submodel descriptor

```
{
  "id": "urn:uuid:da071e28-8cf2-46a1-b5af-65231887c7c5",
  "globalAssetId": "urn:uuid:cdc8b050-ee17-432e-9ca6-b85d8017811d",
  "idShort": "BMW i4 2024-01-01-0001",
  "specificAssetIds": [
    {
      "name": "digitalTwinType",
      "value": "PartInstance"
    },
    {
      "name": "manufacturerId",
      "value": "BPNL0000000J8CWU"
    },
    {
      "name": "manufacturerPartId",
      "value": "G26",
      "externalSubjectId": {
        "type": "ExternalReference",
        "keys": [
          {
            "type": "GlobalReference",
            "value": "PUBLIC_READABLE"
          }
        ]
      }
    },
    {
      "name": "partInstanceId",
      "value": "Skg0S0EzMjcwS0MwMDc0OTc="
    },
    {
      "name": "intrinsicId",
      "value": "Skg0S0EzMjcwS0MwMDc0OTc="
```

```
    },
    {
      "name": "van",
      "value": "Skg0S0EzMjcwS0MwMDc0OTc="
    }
  ],
  "submodelDescriptors": [
    {
      "endpoints": [
        {
          "interface": "SUBMODEL-3.0",
          "protocolInformation": {
            "href": "https://dataplane-partner.edc.aws.bmw.cloud/public/submodel",
            "endpointProtocol": "HTTP",
            "endpointProtocolVersion": [
              "1.1"
            ],
            "subprotocol": "DSP",
            "subprotocolBody": "id=Vehicle_2024-01-01-0001_Submodel_SerialPart;dspEndpoint=https://connector-
partner.edc.aws.bmw.cloud/api/v1/dsp",
            "subprotocolBodyEncoding": "plain",
            "securityAttributes": [
              {
                "type": "NONE",
                "key": "NONE",
                "value": "NONE"
              }
            ]
          }
        }
      ],
      "idShort": "serialPart",
      "id": "urn:uuid:9624aa4d-cfba-466c-b67a-6e7227e3fb54",
      "semanticId": {
        "type": "ExternalReference",
        "keys": [
          {
            "type": "GlobalReference",
            "value": "urn:samm:io.catenax.serial_part:3.0.0#SerialPart"
          }
        ]
      }
    },
    {
      "endpoints": [
        {
          "interface": "SUBMODEL-3.0",
          "protocolInformation": {
            "href": "https://dataplane-partner.edc.aws.bmw.cloud/public/submodel",
            "endpointProtocol": "HTTP",
            "endpointProtocolVersion": [
              "1.1"
            ],
            "subprotocol": "DSP",
            "subprotocolBody": "id=Vehicle_2024-01-01-0001_Submodel_SingleLevelBomAsBuilt;
dspEndpoint=https://connector-partner.edc.aws.bmw.cloud/api/v1/dsp",
            "subprotocolBodyEncoding": "plain",
            "securityAttributes": [
              {
                "type": "NONE",
                "key": "NONE",
                "value": "NONE"
              }
            ]
          }
        }
      ],
      "idShort": "singleLevelBomAsBuilt",
      "id": "urn:uuid:b0a8dca2-34fe-41b4-a19e-710cd337b3b6",
      "semanticId": {
        "type": "ExternalReference",
```

```
        "keys": [
          {
            "type": "GlobalReference",
            "value": "urn:samm:io.catenax.single_level_bom_as_built:3.0.0#SingleLevelBomAsBuilt"
          }
        ]
      }
    }
  ]
}
```

specific Asset ID serialized parts

❓where is external subject id needed?

Ke	Description	external subject Id needed Y/N	Type
mandatory			
manufacturePartId	The ID of the type/catalog part from the *manufacturer*.		
manufacutureId	BPN-L number		
digitalTwinTyp	"PartInstance" for serialized parts, batches, and JIS parts "PartType" for catalog parts		
partInstanceId	The serial number of the part from the manufacturer		
intrinsicId	The serial number of the part from the manufacturer		
optional			
customerPartId	The ID of the type/catalog part from the *customer*		
Van	**Only for vehicles:** The pseudonymized vehicle identification number (VIN) of the vehicle.		

specific Asset ID batch parts

Digital Twin Type : Part Type

Info: Beispiel einer Batteryzelle **LI-ION ZELLE PHEV1 26AH MK02 C2**

href und dsp endpoint fehlen

AAS descirptor + Submodel descriptor

```
{
  "id": "urn:uuid:5d63f272-8029-4ee6-8e3d-3b1831a30b36",
  "globalAssetId": "urn:uuid:67b89e05-7506-4412-a929-1cd548f31861",
  "idShort": "Battery-Cell-Gen5",
  "specificAssetIds": [
    {
      "name": "digitalTwinType",
      "value": "PartType"
    },
    {
      "name": "manufacturerId",
      "value": "BPNL0000000J8CWU"
    },
    {
      "name": "manufacturerPartId",
      "value": "188006",
      "externalSubjectId": {
        "type": "ExternalReference",
        "keys": [
          {
            "type": "GlobalReference",
            "value": "PUBLIC_READABLE"
          }
        ]
```

```
      },
          "name": "customerPartId",
      "value": "2373700",
    }
  ],
  "submodelDescriptors": [
    {
      "endpoints": [
        {
          "interface": "SUBMODEL-3.0",
          "protocolInformation": {
            "href": "https://dataplane-partner.edc.aws.bmw.cloud/public/submodel",
            "endpointProtocol": "HTTP",
            "endpointProtocolVersion": [
              "1.1"
            ],
            "subprotocol": "DSP",
            "subprotocolBody": "id=Submodel_Cell_SecondaryMaterialContent;dspEndpoint=https://connector-partner.
edc.aws.bmw.cloud/api/v1/dsp",
            "subprotocolBodyEncoding": "plain",
            "securityAttributes": [
              {
                "type": "NONE",
                "key": "NONE",
                "value": "NONE"
              }
            ]
          }
        }
      ],
      "idShort": "secondaryMaterialContent",
      "id": "urn:uuid:0815e6b5-418d-4630-bd81-94b322770b5c",
      "semanticId": {
        "type": "ExternalReference",
        "keys": [
          {
            "type": "GlobalReference",
            "value": "urn:samm:io.catenax.secondary_material_content_calculated:1.0.0#SecondaryMaterialContent"
          }
        ]
      }
    },
    {
      "endpoints": [
        {
          "interface": "SUBMODEL-3.0",
          "protocolInformation": {
            "href": "https://dataplane-partner.edc.aws.bmw.cloud/public/submodel",
            "endpointProtocol": "HTTP",
            "endpointProtocolVersion": [
              "1.1"
            ],
            "subprotocol": "DSP",
            "subprotocolBody": "id=Submodel_Cell_ProductCarbonFootprint;dspEndpoint=https://connector-partner.
edc.aws.bmw.cloud/api/v1/dsp",
            "subprotocolBodyEncoding": "plain",
            "securityAttributes": [
              {
                "type": "NONE",
                "key": "NONE",
                "value": "NONE"
              }
            ]
          }
        }
      ],
      "idShort": "productCarbonFootprint",
      "id": "urn:uuid:43c1965b-f86c-409d-a683-9d77ab68b4e8",
      "semanticId": {
        "type": "ExternalReference",
        "keys": [
```

```
          {
            "type": "GlobalReference",
            "value": "urn:samm:com.bmw.pcf:1.0.0#Pcf"
          }
        ]
      }
    }
  ]
}
```

Asset with policies

van als specificAssetId rauslassen, da nur für OEMs wichtig

Setup Submodels of Digital Twins in EDC

For all assets, we use the same access and usage polices. Technically, we create four usage policies, one for SerialPart, one for SingleLevelBomAsBuilt, one for Secondary Material Content and one for Product Carbon Footprint, but the content (except their ID) is exactly the same. You can use different usage policies for submodels, but for SerialPart and SingleLevelBomAsBuilt this is not necessary. The usage purposes are standardized and are stored on the odrl repo of the catena-x association project

Access Policy with BPN Group


```
{
  "@id": "access-policy",
  "@type": "PolicyDefinition",
  "profile": "cx-policy:profile2405",
  "policy": {
    "@type": "odrl:Set",
    "odrl:permission": {
      "odrl:action": {
        "@id": "odrl:use"
      },
      "odrl:constraint": {
        "odrl:and": [
          {
            "odrl:leftOperand": {
              "@id": "cx-policy:Membership"
            },
            "odrl:operator": {
              "@id": "odrl:eq"
            },
            "odrl:rightOperand": "active"
          },
          {
            "odrl:leftOperand": {
              "@id": "tx:BusinessPartnerGroup"
            },
            "odrl:operator": {
              "@id": "odrl:isAnyOf"
            },
            "odrl:rightOperand": "industry-core-example"
          }
        ]
      }
    }
  },
  "@context": {
    "@vocab": "https://w3id.org/edc/v0.0.1/ns/",
    "edc": "https://w3id.org/edc/v0.0.1/ns/",
    "tx": "https://w3id.org/tractusx/v0.0.1/ns/",
    "tx-auth": "https://w3id.org/tractusx/auth/",
    "cx-policy": "https://w3id.org/catenax/policy/",
    "odrl": "http://www.w3.org/ns/odrl/2/"
  }
}
```

Usage Policy for SerialPart Submodels


```
{
  "@id": "usage-policy-for-aspect-model-serialpart",
  "@type": "PolicyDefinition",
  "profile": "cx-policy:profile2405",
  "policy": {
    "@type": "odrl:Set",
    "odrl:permission": {
      "odrl:action": {
        "@id": "odrl:use"
      },
      "odrl:constraint": {
        "odrl:and": [
          {
            "odrl:leftOperand": {
              "@id": "cx-policy:FrameworkAgreement"
            },
            "odrl:operator": {
              "@id": "odrl:eq"
            },
            "odrl:rightOperand": "DataExchangeGovernance:1.0"
          },
          {
            "odrl:leftOperand": {
              "@id": "cx-policy:UsagePurpose"
            },
            "odrl:operator": {
              "@id": "odrl:eq"
            },
            "odrl:rightOperand": "cx.core.industrycore:1"
          }
        ]
      }
    }
  },
  "@context": {
    "@vocab": "https://w3id.org/edc/v0.0.1/ns/",
    "edc": "https://w3id.org/edc/v0.0.1/ns/",
    "tx": "https://w3id.org/tractusx/v0.0.1/ns/",
    "tx-auth": "https://w3id.org/tractusx/auth/",
    "cx-policy": "https://w3id.org/catenax/policy/",
    "odrl": "http://www.w3.org/ns/odrl/2/"
  }
}
```

Contract Definition for All SerialPart Submodels

```
{
  "@id": "Contract-Definition-Vehicle-Submodel-SerialPart",
  "@type": "ContractDefinition",
  "accessPolicyId": "access-policy",
  "contractPolicyId": "usage-policy-for-aspect-model-serialpart",
  "assetsSelector": {
    "@type": "Criterion",
    "operandLeft": "https://admin-shell.io/aas/3/0/HasSemantics/semanticId",
    "operator": "like",
    "operandRight": "%urn:samm:io.catenax.serial_part:3.0.0#SerialPart%"
  },
  "@context": {
    "@vocab": "https://w3id.org/edc/v0.0.1/ns/",
    "edc": "https://w3id.org/edc/v0.0.1/ns/",
    "tx": "https://w3id.org/tractusx/v0.0.1/ns/",
    "tx-auth": "https://w3id.org/tractusx/auth/",
    "cx-policy": "https://w3id.org/catenax/policy/",
    "odrl": "http://www.w3.org/ns/odrl/2/"
  }
}
```

For the above contract definition, be use a filter that is based on one of the properties we use for submodel EDC assets. The filter selects all EDC assets that represent submodels of type SerialPart (aspect model with semanticId: urn:samm:io.catenax.serial_part:3.0.0#SerialPart). With this filter, the contract definition use automatically used for all EDC assets that are correctly created using this property.

Publish Submodels without Asset Bundling (One EDC Asset per Submodel)

For the Vehicle digital twin, we chose an approach without asset bundling, i.e., we create an EDC asset for every submodel of this digital twin. This approach is described as Option 1 in the Industry Core KIT. The Vehicle digital twin "Vehicle 2024-01-01-0001" has two submodels, so we need to create two EDC assets with policies. The necessary steps are the same for both submodels:

- Create the asset for the submodel (SerialPart, SingleLevelBomAsBuilt)
- If not yet created, create access and usage policies for the asset. As these policies might be shared between assets, they only need to be created once.
- Create the contract definition for the asset
- After this step, the EDC asset for the submodel is visible in the EDC catalog for every partner that has access based on the configured access policy in the contract definition.

Asset for Submodel SerialPart for the Vehicle Digital Twin


```
{
  "@context": {
    "dct": "http://purl.org/dc/terms/",
    "cx-common": "https://w3id.org/catenax/ontology/common#",
    "cx-taxo": "https://w3id.org/catenax/taxonomy#",
    "aas-semantics": "https://admin-shell.io/aas/3/0/HasSemantics/"
  },
  "@type": "Asset",
  "@id": "Vehicle_2024-01-01-0001_Submodel_SerialPart",
  "properties": {
    "dct:type": {
      "@id": "cx-taxo:Submodel"
    },
    "cx-common:version": "3.0",
    "aas-semantics:semanticId": {
      "@id": "urn:samm:io.catenax.serial_part:3.0.0#SerialPart"
    }
  },
  "dataAddress": {
    "@type": "DataAddress",
    "secretAccessKey": "...",
    "accessKeyId": "...",
    "region": "eu-west-1",
    "type": "AmazonS3",
    "bucketName": "digital-twin-exchange-example",
    "objectName": "Vehicle_2024-01-01-0001_Submodel_SerialPart.json"
  }
}
```


The asset - via its dataAddress - points to an S3 bucket where the actual submodel is stored as a JSON file. The JSON file for the Vehicle SerialPart submodel looks like this:

Submodel SerialPart for Vehicle Digital Twin


```
{
  "catenaXId": "urn:uuid:cdc8b050-ee17-432e-9ca6-b85d8017811d",
  "localIdentifiers": [
    {
      "key": "manufacturerId",
      "value": "BPNL000000000ISY"
    },
    {
      "key": "partInstanceId",
      "value": "NzFhZmFkNmItMGU5Ny00NzNkLWFiZDItOWY5NWU2NzE0OTVi"
    },
    {
      "key": "van",
      "value": "NzFhZmFkNmItMGU5Ny00NzNkLWFiZDItOWY5NWU2NzE0OTVi"
    }
  ],
  "manufacturingInformation": {
    "date": "2024-01-01T00:00:00.000Z",
    "country": "DEU"
  },
  "partTypeInformation": {
    "manufacturerPartId": "G60",
    "classification": "product",
    "nameAtManufacturer": "Vehicle"
  }
}
```

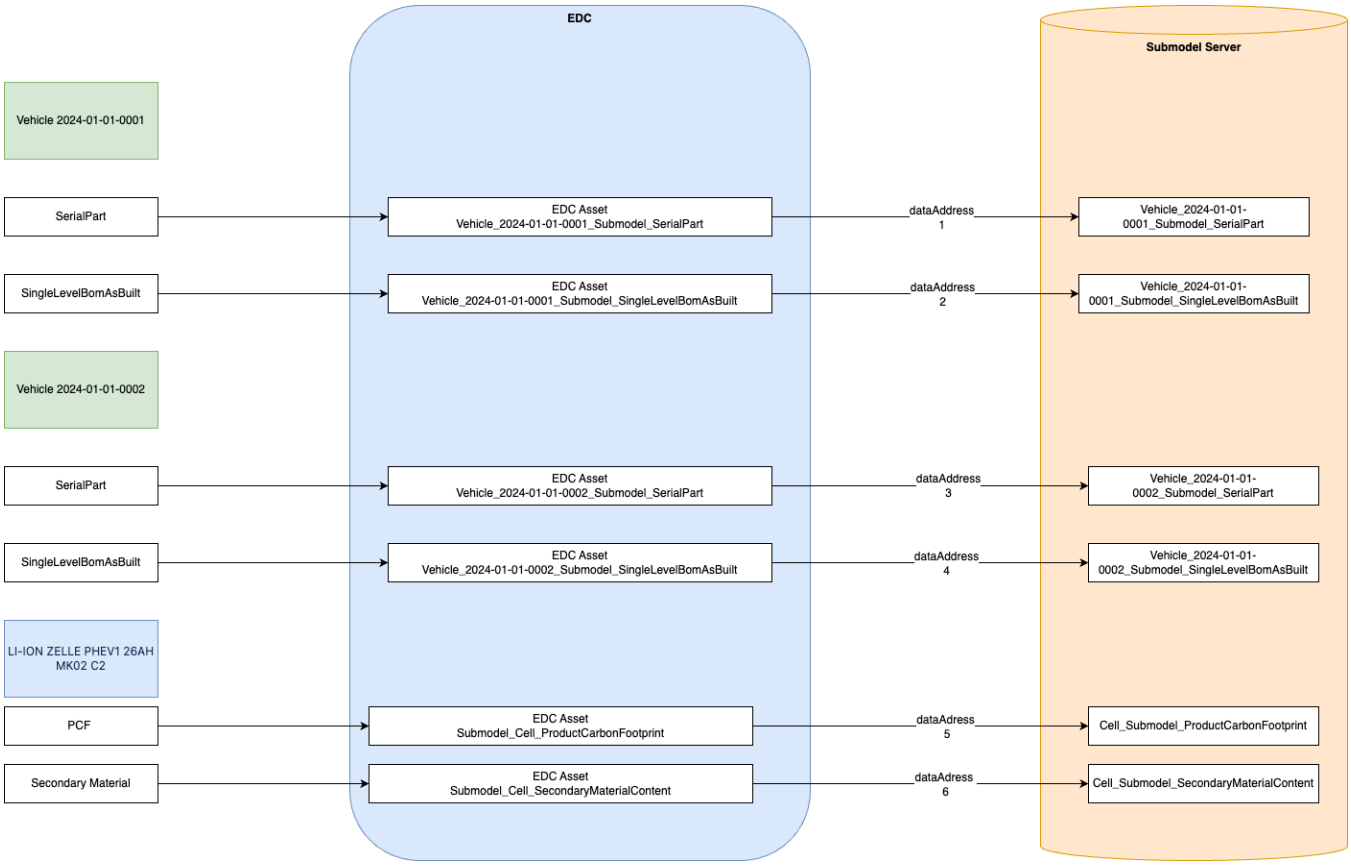

The digital twins for Vehicle do not use asset bundling, but one asset is created for every submodel of a Vehicle digital twin:

- The actual submodel is represented by the asset in the EDC
- href is only the URL to the EDC data plane

Use the same usage policy for all serialized submodels (SerialPart and SingleLevelBomAsBuilt)?

Use the same usage policy for all type based submodels

On a more general level, when you create several Vehicle digital twins, the structure should look like in the following image:



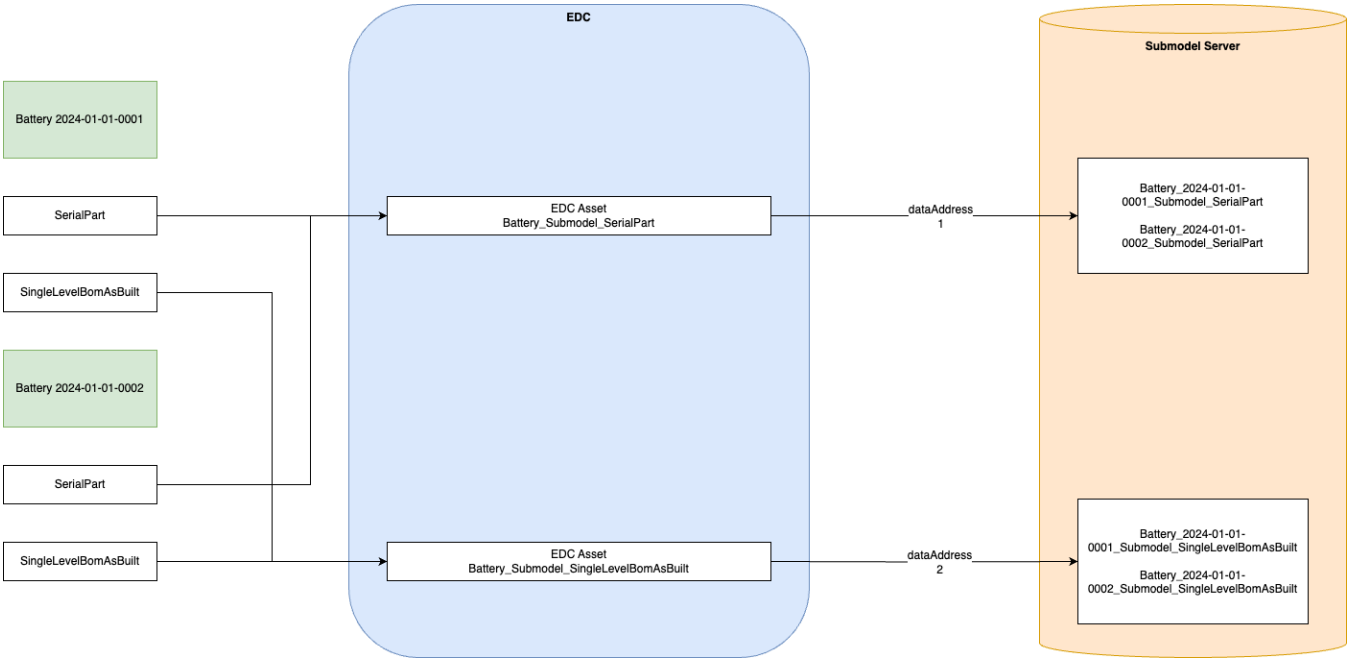
Publish Submodels with Asset Bundling (One EDC Asset per Aspect Model)

For the Battery digital twin, we chose an approach with asset bundling, i.e., we create an EDC asset for every aspect model that is used to transfer all submodels of this aspect model for *all* Battery digital twin. This approach is described as Option 2 in the Industry Core KIT. The Battery digital twin "Battery 2024-01-01-0001" only has one submodel, so we only need to create one EDC assets with policies.

- Create the asset for the aspect model of Battery digital twins (SerialPart)
- If not yet created, create access and usage policies for the asset. As these policies might be shared between assets, they only need to be created once.
- Create the contract definition for the asset
- After this step, the EDC asset for SerialPart submodels of Battery digital twins is visible in the EDC catalog for every partner that has access based on the configured access policy in the contract definition.

Asset for Submodel SerialPart for the Battery Digital Twin

```
{
  "@context": {
    "dct": "http://purl.org/dc/terms/",
    "cx-common": "https://w3id.org/catenax/ontology/common#",
    "cx-taxo": "https://w3id.org/catenax/taxonomy#",
    "aas-semantics": "https://admin-shell.io/aas/3/0/HasSemantics/"
  },
  "@type": "Asset",
  "@id": "Battery_Submodel_SerialPart",
  "properties": {
    "dct:type": {
      "@id": "cx-taxo:Submodel"
    },
    "cx-common:version": "3.0",
    "aas-semantics:semanticId": {
      "@id": "urn:samm:io.catenax.serial_part:3.0.0#SerialPart"
    }
  },
  "dataAddress": {
    "@type": "DataAddress",
    "type": "HttpData",
    "baseUrl": "https://c132ew588i.execute-api.eu-west-1.amazonaws.com/dev/digital-twin-exchange-example",
    "proxyQueryParams": "false",
    "proxyPath": "true",
    "proxyMethod": "false",
    "authKey": "x-api-key",
    "authCode": "..."
  }
}
```

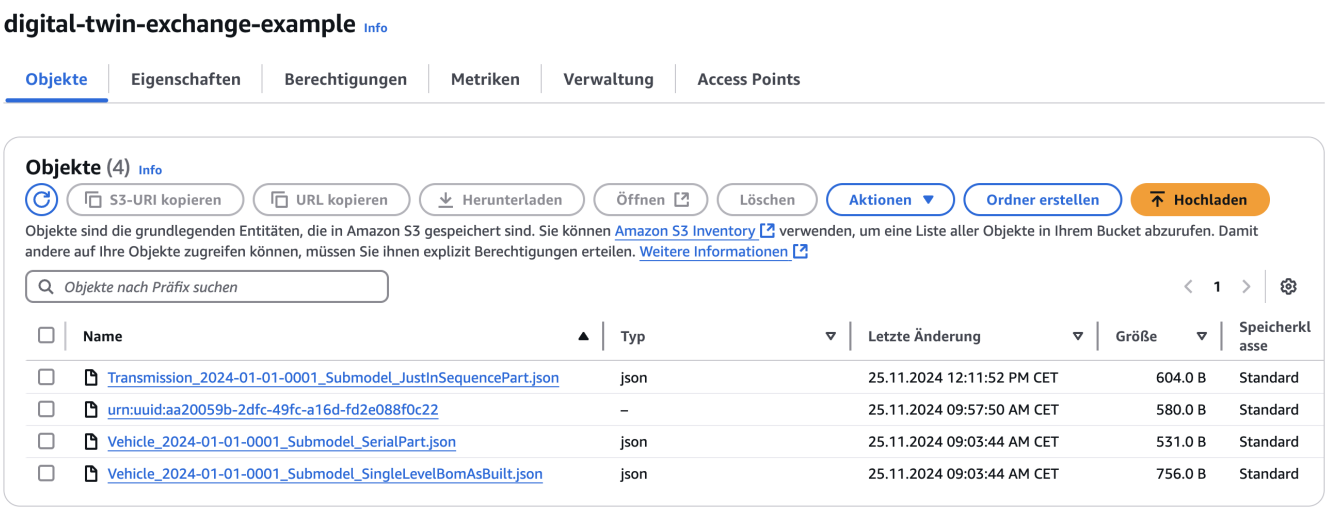


The asset bundling used in this example is a simplified version. With asset bundling, you need to ensure in the Submodel Server, that the data consumer, that transfers the data via the EDC, is actually allowed to see and transfer the submodel for a digital twin, if there are more than one partner allowed to fetch, e.g, Battery submodels. One way to solve this is to create separate EDC assets per supplier (and per submodel), so, e.g. Battery_Submodel_SerialPart_Partner_A and Battery_Submodel_SerialPart_Partner_B.

Create Submodels in the Submodel Server

For this example, we use a very simple Submodel Server. Basically, the Submodel Server is referenced in the `dataAddress` property of the EDC asset description. The actual properties and values depend on the Submodel Server you use. So, this documentation can only serve as an example.

For non-asset-bundling submodels, we use a simple S3 bucket as Submodel Server:



The `dataAddress` in the EDC asset directly points to the submodel (JSON file) in the S3 bucket that will be transferred when a data consumer requests this submodel:

```
 "dataAddress": {
  "@type": "DataAddress",
  "secretAccessKey": "...",
  "accessKeyId": "...",
  "region": "eu-west-1",
  "type": "AmazonS3",
  "bucketName": "digital-twin-exchange-example",
  "objectName": "Vehicle_2024-01-01-0001_Submodel_SerialPart.json"
}
```

Without asset bundling, every submodel points to a unique EDC asset which references a unique JSON file in the S3 bucket. So, it's easily extendible with further submodels for Vehicle digital twins, e.g., with Vehicle_2024-01-01-0002_Submodel_SerialPart.json, Vehicle_2024-01-01-0003_Submodel_SerialPart.json, and so o.

For asset-bundling submodels, we cannot do that as the same EDC asset must be able to return not just one fixed submodel, but a submodel that depends on the input request to the EDC / Submodel Server. Therefore, we use a (simple) REST API to serve these requests. The `dataAddress` in this case looks like this:

```
 "dataAddress": {
  "@type": "DataAddress",
  "type": "HttpData",
  "baseUrl": "https://c132ew588i.execute-api.eu-west-1.amazonaws.com/dev/digital-twin-exchange-example",
  "proxyQueryParams": "false",
  "proxyPath": "true",
  "proxyMethod": "false",
  "authKey": "x-api-key",
  "authCode": "..."
}
```

After a successful contract negotiation, a data consumer invokes the data plane of the EDC with the URL specified in the `href` property of the submodel descriptor. For the Battery digital twin, the `href` looks like this:

```
https://dataplane-partner.edc.aws.bmw.cloud/public/urn:uuid:aa20059b-2dfc-49fc-a16d-fd2e088f0c22/submodel
```

Note that the data consumer must appen "/$value" to this URL as defined in the Digital Twin KIT.

The EDC will transform this URL given the information in the asset's `dataAddress` by replacing the EDC dataplane with the `basePath` from the `dataAddress`.

```
https://c132ew588i.execute-api.eu-west-1.amazonaws.com/dev/digital-twin-exchange-example/urn:uuid:aa20059b-2dfc-49fc-a16d-fd2e088f0c22/submodel/$value
```

Then, the EDC retrieves the data from this URL using a GET request and returns the data.

We implemented a simple REST API that retrieves the submodel specified in the URL (urn:uuid:aa20059b-2dfc-49fc-a16d-fd2e088f0c22) from the above S3 bucket.

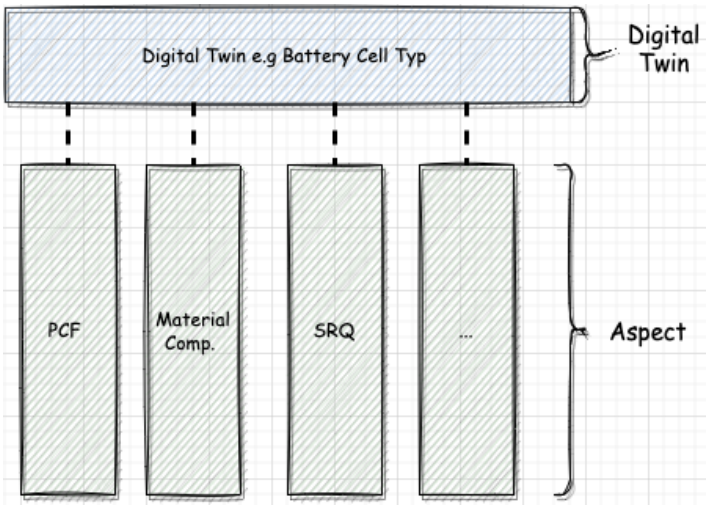
The difference to non-asset bundling can be seen when comparing both href URLs:

- Without asset-bundling: https://dataplane-partner.edc.aws.bmw.cloud/public/submodel/$value
  - Here, the EDC asset directly points to the submodel JSON file (via its dataAddress), so no further information besides the dataplane is needed in the href URL.
- With asset-bundling: https://dataplane-partner.edc.aws.bmw.cloud/public/urn:uuid:aa20059b-2dfc-49fc-a16d-fd2e088f0c22/submodel/$value
  - Here, the EDC asset is shared between all submodels of the SerialPart aspect model for Battery digital twins. Without a URL like the one without asset bundling, the Submodel Server (i.e., the REST API) would not know which of all available submodels it should return. Therefore, the href URL must specify which submodel actually to fetch based on the submodel descriptor (and therefore digital twin) in which its defined.
  - For the example, we use the submodel ID as ID as it's must be unique anyway. The REST API Submodel Server uses this submodel ID and looks for the corresponding file in the S3 bucket shown above.

Use Full Definition

Aspect	Aspects are services that provide data about assets represented by Digital Twins. Each aspect exposes an endpoint for supported data transport protocols (HTTP or MQTT-> Not used at the moment in Catena-X). When a data consumer, typically a service that intends to use the data, locates a relevant Twin in the Digital Twin Registry, it can obtain two key pieces of information from that Twin: 1. The endpoint of the aspect 2. The identifier of the Aspect Model corresponding to the aspect's runtime data The data consumer can then retrieve data from the aspect as needed. The aspect implementer is responsible for ensuring the runtime data sent by the aspect aligns with the respective Aspect Model. Aspect data consumers can be implemented in two ways: 1. Specifically for a particular aspect type, by referencing the corresponding Aspect Model during development or generating client/deserialization code from it. 2. Generically, by interpreting the Aspect Model at runtime and using that to deserialize the data.	N/A	

Aspect Model	An Aspect Model is a formal (machine-readable) description of the structure and semantics of the data provided by an Aspect. The Aspect Model serves two key purposes: 1. Capturing Domain Semantics: • The Aspect Model captures the domain-specific semantics of the part of the Digital Twin it describes, acting as an ontology. • It makes explicit and consistent the information that is often available informally (e.g., in a data sheet) or implicitly (as tacit knowledge) across a broader scope than a single service or application. 2. Defining a Contract: • The Aspect Model serves as a contract between the data provider and data consumer, similar to a schema description (e.g., JSON Schema, XML Schema). • It precisely defines the data structures and values that may appear in the runtime data. • Unlike a pure schema language, the Aspect Model also contains the previously mentioned domain semantics, which can be used for data validation beyond just the structural constraints. In summary, the Aspect Model bridges the gap between the domain-specific knowledge and the technical representation of data, providing a formal, machine-readable way to describe and validate the information contained within a Digital Twin's Aspect.	**Aspect Model:** ``` :BatteryPass a samm:Aspect ; samm:preferredName "Battery Passport"@en ; samm:description "The battery pass describes information collected during the lifecycle of a battery. samm:see <https://environment.ec.europa.eu/publications/proposal-ecodesign-sustainable-products-regula samm:see <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32023R1542&qid=1693215264963> ; samm:properties (:identification :operation :characteristics :sustainability :materials :performance samm:operations ( ) ; samm:events ( ) . :identification a samm:Property ; samm:preferredName "Identification"@en ; samm:description "Battery identification involves specific battery identifiers, including those assig samm:characteristic :IdentificationCharacteristic . :operation a samm:Property ; samm:preferredName "Operation"@en ; samm:description "General operation information includes details such as the manufacturer's identifica samm:characteristic :OperationCharacteristic . ``` **Aspect Model Data:** ``` { "characteristics" : { "physicalDimension" : { "length" : { "value" : 20.0, "unit" : "unit:millimetre" }, "width" : { "value" : 20.0, "unit" : "unit:millimetre" }, "weight" : { "value" : 20.0, "unit" : "unit:gram" }, "height" : { "value" : 20.0, "unit" : "unit:millimetre" } }, "warranty" : { "lifeValue" : 36, "lifeUnit" : "unit:day" } }, "metadata" : { "backupReference" : "https://dummy.link", "registrationIdentifier" : "https://dummy.link/ID8283746239078", "economicOperatorId" : "BPNL0123456789ZZ", "lastModification" : "2000-01-01", "predecessor" : "urn:uuid:00000000-0000-0000-0000-000000000000", "issueDate" : "2000-01-01", "version" : "1.0.0", "passportIdentifier" : "urn:uuid:550e8400-e29b-41d4-a716-446655440000", "status" : "draft", "expirationDate" : "2030-01-01" }, "commercial" : { "placedOnMarket" : "2000-01-01", "purpose" : ["automotive"] }, "identification" : { "chemistry" : "Nickel Cobalt Manganese (NCM)", "idDmc" : "34567890", "identification" : { "batch" : [{ "value" : "BID12345678", "key" : "batchId" }], "codes" : [{ "value" : "8703 24 10 00", "key" : "TARIC" }], "type" : { "manufacturerPartId" : "123-0.740-3434-A", "nameAtManufacturer" : "Mirror left" } ``` 

Asset Administration Shell	The Asset Administration Shell is a standardized framework that enables cross-vendor interoperability and supports the digital twin concept for Industry 4.0. Key features: • Enables standardized data exchange across product life cycles and production processes • Applicable to both intelligent and non-intelligent products • Provides a digital and interoperable foundation for autonomous systems and AI • Structures product data into standardized submodels for efficient processing and access • Supports the mapping of entire product life cycles and continuous value chains • Can present implementation challenges, especially for SMEs, that require guidance and support	N/A	
Digital Twins	A Digital Twin is a digital representation of a physical asset, such as a machine, vehicle, or product. It can also represent a virtual or logical entity, like a machine type, product type, server, or fact sheet. There are multiple ways to implement the concept of a Digital Twin in practice. These include: 1. Representing a physical asset: The Digital Twin can mirror a real-world physical object, capturing its state, properties, and behavior. 2. Representing a virtual/logical entity: The Digital Twin can model a virtual or logical thing, such as a product type or server configuration. 3. Hybrid approach: The Digital Twin can combine elements of both physical and virtual/logical representations. The specific implementation approach depends on the use case and the nature of the asset or entity being modeled. Regardless of the approach, the core idea of a Digital Twin is to create a digital counterpart that can be used to simulate, analyze, and optimize the physical or virtual original.		
Sub Model	A submodel is used to structure the information model of an AAS (Asset Administration Shell) representing the technical functionality of the underlying asset into distinguishable parts. Each submodel refers to a well-defined domain or subject matter. Submodels can become standardized and thus become submodel templates. Submodels can have different life-cycles.	See Digital Twin Registry	

Digital Twin Registry	The Digital Twin Registry (DTR) serves as the central point of information about the existence of: • Digital Twins and their associated Aspects • Asset Administration Shells and their Submodels The Digital Twin Registry is the primary means to locate and interact with digital twins. To enable this, the Digital Twin Registry provides several APIs. Through these interfaces applications can discover, access, and work with the digital twins and their related components stored in the registry.The Digital Twin Registry acts as a hub, allowing users to find the necessary information and entry points to engage with the digital representations of physical or virtual assets.